

Differential Equations

Dynamics of Oscillators & Linear Systems

A Comprehensive Guide to 2nd-Order Equations, The Superposition Principle, and The Mass-Spring Model.

Part I

2nd Order Homogeneous Equations

A second-order linear homogeneous ODE with constant coefficients takes the general form:

$$ay'' + by' + cy = 0$$

- ✓ **Linearity:** The dependent variable y and its derivatives appear only to the first power.
- ✓ **Homogeneity:** The right side is zero, meaning no external "forcing" is applied.
- ✓ **Constant Coefficients:** a , b , and c are real numbers.

The Characteristic Equation

Assuming a solution of the form $y = e^{rt}$, we derive the quadratic equation:

$$ar^2 + br + c = 0$$

The roots (r) determine the fundamental nature of the system's behavior.

The Superposition Principle

Theoretical Statement

If $y_1(t)$ and $y_2(t)$ are solutions to a homogeneous linear ODE, then any **linear combination** is also a solution:

$$y(t) = c_1 y_1 + c_2 y_2$$

This allows us to construct the **General Solution** from a fundamental set of basic solutions.

Proof Sketch

Let $L[y] = ay'' + by' + cy$. By linearity:

$$L[c_1 y_1 + c_2 y_2] = c_1 L[y_1] + c_2 L[y_2]$$

Since $L[y_1] = 0$ and $L[y_2] = 0$, the sum must be zero. ✓

Part II

Modeling the Physics

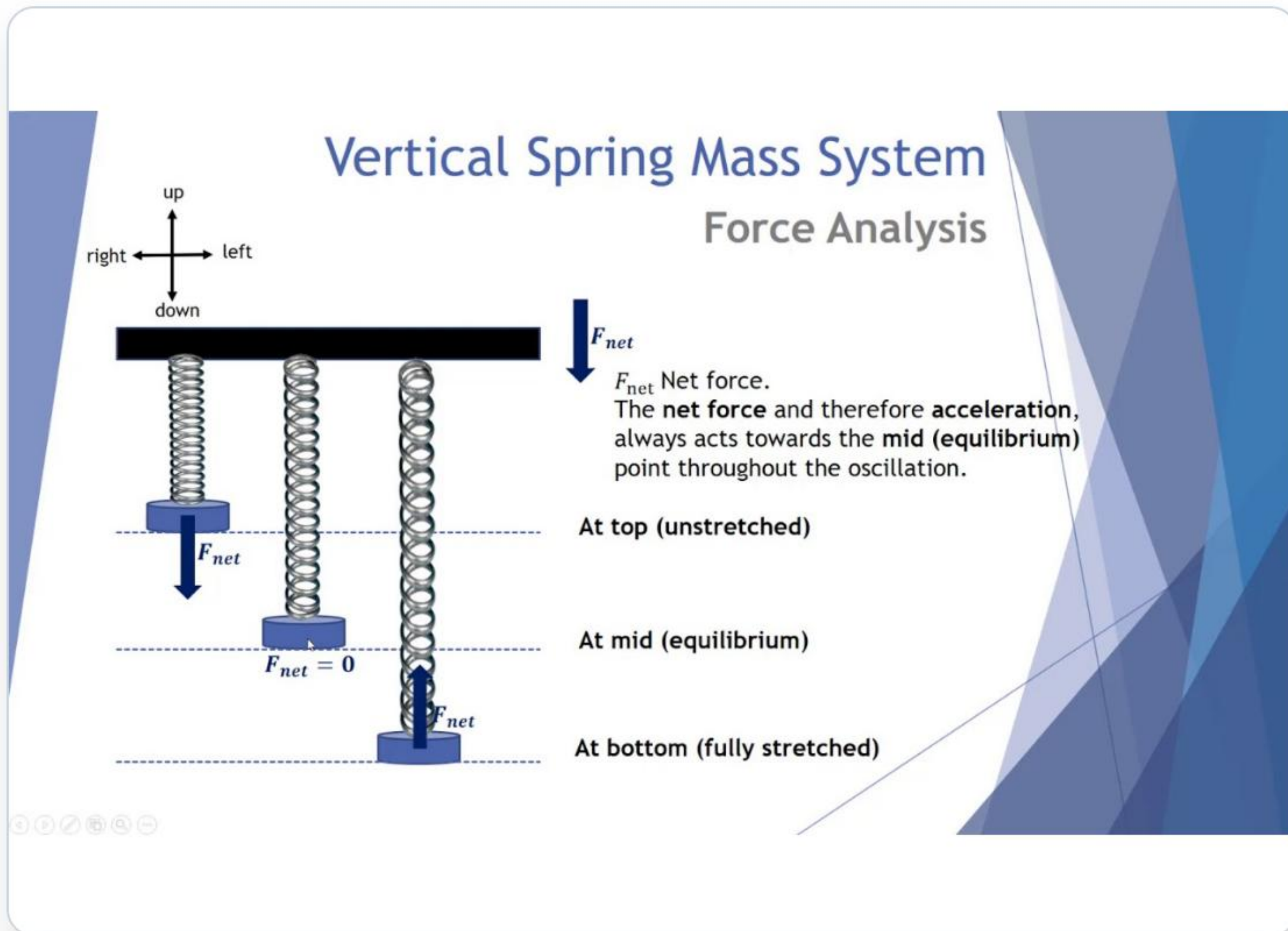


Figure: Mass m on a spring with constant k in equilibrium.

We combine **Newton's 2nd Law** with **Hooke's Law** and **Friction (Damping)**:

- ✓ **Inertia:** $F = ma = m\ddot{x}$
- ✓ **Restoring Force:** $F_s = -kx$
- ✓ **Damping Force:** $F_d = -b\dot{x}$

$$m\ddot{x} + b\dot{x} + kx = f(t)$$

Where $f(t)$ is the external driving force.

The Three Damping States

1. Underdamped

Roots are complex: $r = \alpha \pm i\beta$.

The system oscillates with a decaying amplitude. Most common in physical watches or vibrations.

2. Overdamped

Roots are real and distinct: $r_1 \neq r_2$.

High friction. The mass crawls back to equilibrium without any oscillation.

3. Critically Damped

Repeated real root: $r_1 = r_2$.

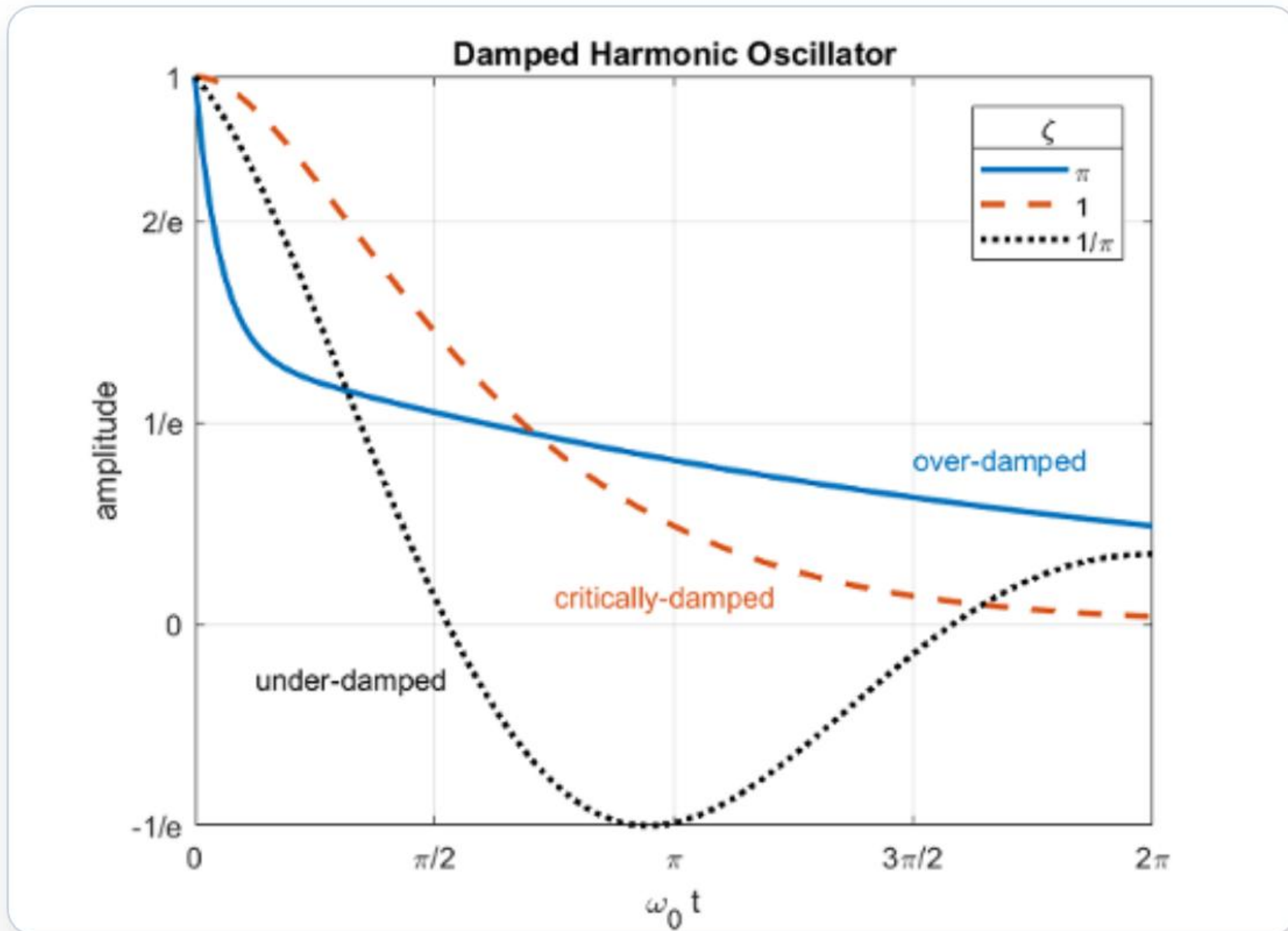
The fastest return to equilibrium without "overshoot."
Ideal for car suspension systems.

Undamped

Pure imaginary roots: $r = \pm i\omega$.

Simple Harmonic Motion (SHM). Energy is conserved; oscillation continues forever.

Visualizing Harmonic Motion



Intuition: The Root Location

- ✓ **Real Part:** Represents **Decay**. If negative, system settles. If zero, system oscillates perpetually.
- ✓ **Imaginary Part:** Represents **Frequency**. The "wiggle" of the system.
- ✓ **Phase Space:** Plotting Velocity vs. Position shows spirals (damping) or circles (undamped).

Part III

Method of Undetermined Coefficients

Used to find a **Particular Solution** (y_p) when the driving force $f(t)$ is a simple function.

Forcing Function $f(t)$	Standard Guess for y_p
Polynomial: $3t^2 + 1$	$At^2 + Bt + C$
Exponential: $5e^{3t}$	Ae^{3t}
Sinusoidal: $\sin(2t)$ or $\cos(2t)$	$A\cos(2t) + B\sin(2t)$

The Adjustment Rule: If the guess matches a term in the homogeneous solution (y_h), multiply your guess by t (or t^2) until it is linearly independent.

The Complete Solution

Y

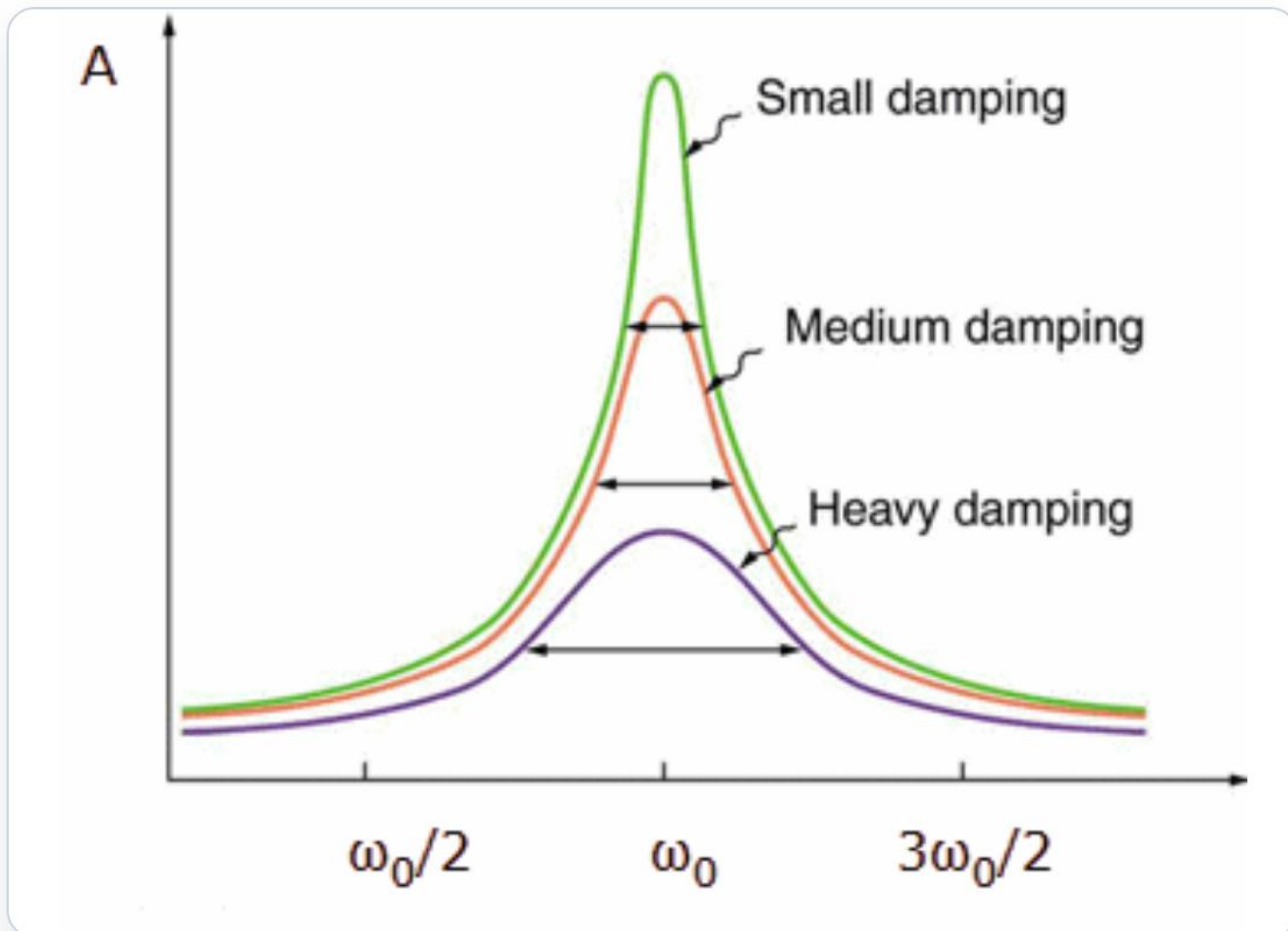
$$= y_h + y_p$$

The Composite Strategy

- ✓ **Step 1:** Solve the homogeneous equation ($f(t) = 0$) to get the "transient" behavior.
- ✓ **Step 2:** Use Undetermined Coefficients to find the "steady-state" response to the forcing.
- ✓ **Step 3:** Add them together. The transients die out over time, leaving only the steady response.

Resonance: The Power of Forcing

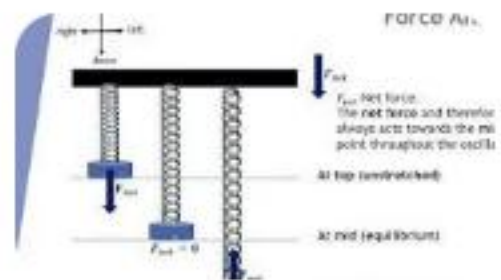
Resonance occurs when the frequency of the external driver matches the **natural frequency** of the system.



Summary & Study Tips

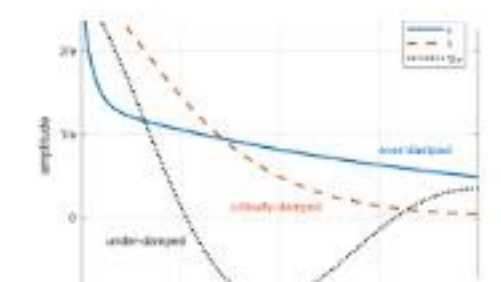
- ✓ **Oscillator:** Remember $mx'' + bx' + kx = f$.
- ✓ **Superposition:** Linearity is the bedrock of engineering. Summing solutions works because of it.
- ✓ **Coefficient Method:** Always check y_h first to avoid "resonance" in your math!
- ✓ **Intuition:** Math describes physics. If your roots are negative, your system is settling. If complex, it's wiggling!

Image Sources



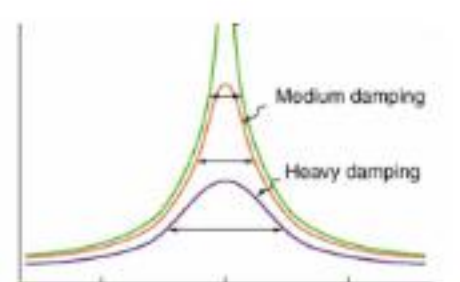
<https://i.ytimg.com/vi/nQSpqYMksoM/maxresdefault.jpg>

Source: www.youtube.com



https://fr.mathworks.com/help/examples/symbolic/win64/ThePhysicsOfTheDampedHarmonicOscillatorExample_01.png

Source: fr.mathworks.com



<http://labman.phys.utk.edu/phys221core/modules/m11/images/amp.gif>

Source: labman.phys.utk.edu